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An Activity-Based Sleep Monitor System for Ambulatory Use

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Summary: Wrist activity measured with a piezoceramic transducer was digitized and analyzed together with subjects' sleep/wake status to derive an optimal method for automatic computer sleep/wake scoring. Several algorithms for quantifying periods of activity were considered, and an algorithm that summed changes in activity level over a 2-s interval was found most sensitive. A computer program for scoring sleep/wake from the resulting digital activity records was then developed, and parameters derived by comparison with subjects' sleep/wake status as determined by EEG. EEG and activity sleep/wake scores agreed 94.46% of the time. A further prospective test of the automatic scoring system with new data yielded agreement of 96.02%. Finally, the data collection and recording functions were implemented in a wearable microprocessor-based digital activity monitor. The automatic scoring program was adjusted to use activity data collected by this monitor, and agreed 93.88% with EEG scoring. A prospective test with new data agreed 93.04% with EEG. Automatic scoring of activity data for sleep/wake is not only fast and accurate, but allows sleep to be monitored in non-laboratory situations. In addition, the score is objective and reliable, and free of scorer bias and drift. **Key Words:** Sleep—Sleep estimation—Wrist activity—Computer sleep scoring—Digital activity recording.

Mullaney et al. (1) have shown that wrist activity recordings can be scored for sleep/wake with accuracy approaching that of EEG scoring. In that study, simultaneous EEG, EOG, EMG, and wrist activity data were collected from 102 subjects. Separate raters scored the EEG-EOG-EMG records and activity records for sleep/wake. The raters agreed 94.5% of the time, only slightly less than the agreement of two raters scoring the same EEG record. Mullaney et al. (1) also noted that the expense in time and effort, as well as the skill, involved in obtaining and scoring activity data is a fraction of that required for scoring EEG recordings. The activity transducer is more comfortable and convenient to wear, and thus activity recording is more adaptable to non-laboratory situations. Best of all, the essential

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information in activity data is quantitative—the amount of activity in a unit of time—and as such potentially adaptable to automatic scoring by computer. This article describes the development of an automatic sleep/wake scoring system and its implementation in a digital sleep monitor capable of recording ambulatory subjects. A more comprehensive version of this report, including tables of data for individual subjects, is also available (2).

EXPERIMENT I

The first requirement in developing an activity-based sleep/wake scoring system is to describe a period of continuous activity information quantitatively. While translating an analog voltage into a digital value is easily accomplished with an analog-to-digital (A/D) converter, the period of time to be considered (the data epoch), the A/D conversion rate, and an algorithm for translating a sequence of A/D conversions into a measure of activity for each data epoch must all be chosen. It is also desirable to filter out 60-Hz electrical interference at the A/D conversion stage. This interference may be caused by an electric blanket or clock at the bedside, and might be interpreted as activity if not recognized and rejected.

Since the goal of this effort was to identify sleep/wake in 1-min scoring intervals, the data epoch could not be greater than 1 min. Shorter data epochs would require storage and management of more data and would limit the length of a recording session when memory capacity is limited. For this study, a 2-s data epoch was adopted as a compromise between a need for detail and the limits of practicality.

The A/D conversion rate is similarly constrained by a need to sample the continuous analog voltage frequently enough to reflect accurately the modulations of the signal, but not so frequently as to overload the system with superfluous information. The conversion rate selected for this study was also selected to permit rejection of 60-Hz electrical interference. N samples of a sinusoid of frequency F when sampled at a rate of $N \times F$ (where N is an even integer) will sum to zero. Summing four A/D conversions at 60×4 or 240 Hz will cancel any 60-Hz component of an actigraph signal and also any 120-Hz harmonic. An A/D conversion rate of 240 Hz was therefore adopted for this study.

With the sum of every four conversions at 240 Hz serving as an artifact-free measure of activity, the remaining question was how to treat the 120 such sums in each 2-s data epoch to represent as well as possible the activity occurring in that epoch for sleep scoring purposes. Experiment I compared the ability of an automatic sleep scoring program to score sleep/wake from analog activity records, using 10 alternative data compression algorithms.

Methods

Signals from an actigraph worn on a watchband were recorded on cassette tapes with a belt-worn Medilog[1] recorder (3). Subjects were instructed to place the actigraph on their dominant wrist in the early evening, wear it all night, and take it off some time after waking in the morning.

[1] Ambulatory Monitoring, Inc., 731 Saw Mill River Road, Ardsley, NY 10502.

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The procedure for processing the recorded data is illustrated in block diagram form in Fig. 1. The activity analog was input to one channel of a Grass Model 78 polygraph and, at the same time, to the computer's A/D converter. Simultaneously, the computer supplied a synchronization time code to a second channel of the polygraph to facilitate scoring.

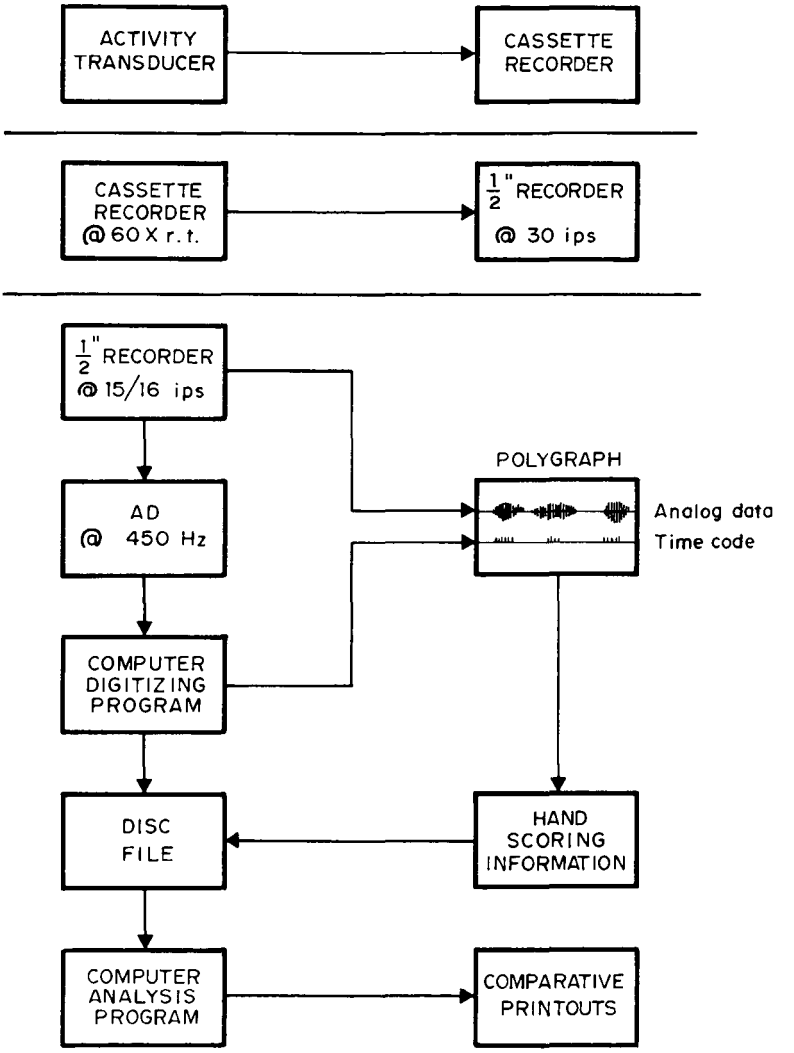


FIG. 1. Block diagram of procedure used to digitize activity records in Experiment I. Output of an actigraph was recorded on cassette tape. The cassette was later replayed at 60 times recording speed to 1/2-inch tape. The 1/2-inch tape was then replayed at 1/32 recording speed to a polygraph and through an analog-to-digital (AD) converter to a computer. The computer generated and stored in disc files 10 digital transformations of the analog data, and also wrote a time code onto the polygraph record. After the polygraph activity was scored for sleep/wake, this information was added to the disc files. This combined activity and sleep/wake data were then examined to arrive at an optimal algorithm for digitizing activity data for sleep scoring purposes.

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After summing every four conversions (Σy) and squared conversions (Σy^2), 10 transformations of Σy and Σy^2 were accumulated over each 2 s and the resulting 10 digital values stored on disc. The 10 algorithms were: the sum of the sums $\Sigma(\Sigma y)$; the sum of the squared sums $\Sigma(\Sigma y^2)$; the sum of the sums squared $\Sigma(\Sigma y)^2$; the sum of the squared sums squared $\Sigma(\Sigma y^2)^2$. Four other algorithms computed a *difference score* equal to 10 times the value of a given item less the value of the preceding and following five items:

$$f[X(i)] = 10 \times X(i) - [X(i-5) + X(i-4) + X(i-3) + X(i-2) + X(i-1) + X(i+1) + X(i+2) + X(i+3) + X(i+4) + X(i+5)]$$

Algorithms 5–8 replaced X in the above expression with (5) Σy, (6) Σy^2, (7) $(\Sigma y)^2$, and (8) $(\Sigma y^2)^2$. Algorithm 9 counted the number of Σys exceeding 90% of the maximum Σy, and algorithm 10 counted the number of Σy^2s exceeding 90% of the maximum Σy^2 every 2 s.

Seven overnight wrist actigraph recordings were processed by the procedure described above. Portions of the records were displayed on a plotter to allow a visual examination of the output of the 10 algorithms during different forms of activity. A more rigorous analysis of the adequacy of each algorithm was obtained by first visually scoring each of the seven polygraph activity records for sleep/wake. An automatic sleep scoring program then scored the digital activity data for sleep/wake and determined the best agreement between computed and visually scored sleep/wake status. The sleep scoring program scored a period of 1 min as wake if the activity value in X of the 30 epochs exceeded a threshold of Y. The X, Y parameter space was searched and the maximum agreement determined for each record and each algorithm. The thresholds producing the best agreement were calculated individually for each record, and therefore differed from record to record. Since in practice a sleep scoring program would have to discriminate sleep from wake by use of the same threshold for all records, a second parameter search was conducted using a single threshold for all records taken together.

Results

Figure 2 shows the computer display and polygraph writeout of a 5-min portion of a record contaminated with 60-Hz noise from an electric blanket. The 10 horizontal traces on the plot represent the 10 digital transformations of the analog activity displayed on the polygraph. Of particular interest in this figure is the contrast between traces 1, 3, 5, and 7 and traces 2, 4, 6, and 8 during periods of electric blanket noise. Since these latter transformations square A/D conversions prior to summation, all values are positive and cancellation of 60-Hz noise cannot occur. Algorithms 1, 3, 5, and 7 do not square conversions, and cancellations of noise can and do occur. The absence of noise in these latter traces indicates the effectiveness of the simple digital filtering technique.

The potential of each of the 10 digital data compression algorithms was evaluated by calculating the maximum agreement between sleep/wake status computed by use of each of the 10 digital activity records and the visually scored actigraph records. When each digital record was scored using the threshold found best for

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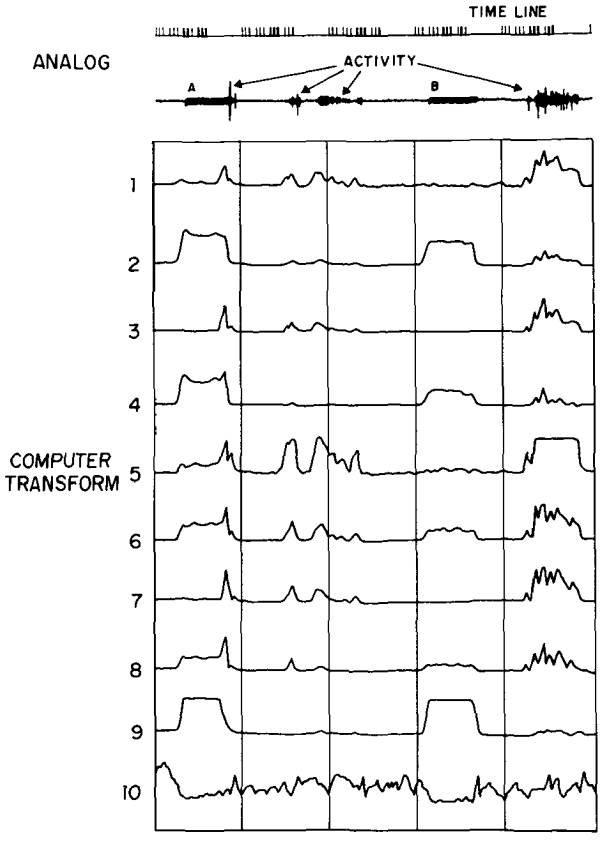


FIG. 2. A 5-min portion of analog polygraph record (*top*) and corresponding computer plotter display. Polygraph record includes a time code and shows activity mixed with 60-Hz interference from an electric blanket (A and B). Computer display plots the 10 digital transformations of the activity and interference data. Transformations 1, 3, 5, 7 effectively remove the interference.

that individual record, algorithms 5 and 6 (difference scores) were superior to the alternative algorithms. When all records were scored using the single threshold found best over all records, algorithm 5 (difference score using Σy) emerged with the best overall agreement.

EXPERIMENT II

Developing an effective technique for digitizing analog actigraph data greatly advanced the prospect of a fully automatic activity-based sleep scoring system. The algorithm selected in Experiment I generated digital activity records which could be retrospectively scored automatically with over 90% agreement with visually scored activity records. It should be noted, however, that Experiment I was concerned with the relative effectiveness of the 10 digital data compression algorithms, and not with the absolute accuracy of the scoring system. The maximum agreement reported in each case represented the best score obtainable retrospectively, that is, with knowledge of the *true* score, and was not a prospective test of a scoring system. Furthermore, the true score was based on visual scoring of actigraph data and only approximated the accepted EEG scoring standard. Fi-

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nally, the automatic sleep scoring algorithm used in Experiment I was also chosen to select between the alternative data compression algorithms, and simply assumed which properties of the record best reflect sleep and wake. In Experiment II, an algorithm for automatically scoring sleep/wake from digitized activity data was developed empirically by selecting parameters that improved agreement with EEG sleep/wake scores.

Methods

Data were obtained from subjects participating in studies involving EEG recording during both wakefulness and sleep. The wrist activity transducer signal was sampled by the computer's A/D converter at a conversion rate of 240 Hz. The analog data were digitized and stored as in Experiment I, but only the transformation selected in Experiment I (#5) was used in data analysis. A total of 20 records (13,488 min) was collected from 17 subjects (3 subjects were recorded twice).

Development of the sleep recognition algorithm began with an expression incorporating a weighted sum of combinations of the digital data with potential for discriminating sleep from wakefulness. Specifically, the expression took the form:

$$D = S \times [W(1)T(1) + W(2)T(2) + W(3)T(3) + W(4)T(4) + W(5)T(5) + W(6)T(6)]$$

where S is a scale factor; W is a weight; $T(1)$ is the sum of the digital activity values for all 30 2-s data epochs in 1 min; $T(2)$ is the activity value for the single most active epoch; $T(3)$ is the sum of the activity values in the two most active epochs separated by at least 30 s; and $T(4)$ is the sum of the activity values in the 8 most active epochs.

Terms $T(5)$ and $T(6)$ were themselves weighted sums of term $T(1)$ over the preceding 4 and following 2 min:

$$T(5) = W(5,1)T(i-1) + W(5,2)T(i-2) + W(5,3)T(i-3) + W(5,4)T(i-4)$$

$$T(6) = W(6,1)T(i+1) + W(6,2)T(i+2)$$

where $T(i-1)$ is the value of $T(1)$ for the preceding minute, $T(i+1)$ for the following minute, etc.

A minute was scored *wake* if $D \geq 1.0$. For each given combination of weights, a range of scale factors was substituted into the above expression for each minute, and the resulting sleep/wake scores compared with the EEG sleep/wake score for that minute. The proportion of minutes for which the automatic score and EEG score agreed was then computed for each scale value, and the maximum agreement served as a retrospective measure of the effectiveness of the weighting. The computer program varied the weighting of one term at a time, and searched for the combination of weights that produced the highest agreement.

As preliminary results became available, it became apparent that better agreement was obtained when $W(1) = W(3) = W(4) = 0$, i.e., the maximal epoch value in each minute was the best discriminator of sleep and wake. Accordingly, a second expression was developed:

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$$D = S \times [W(1)T(i-4) + W(2)T(i-3) + W(3)T(i-2) + W(4)T(i-1)] + W(5)T(i) + W(6)T(i+1) + W(7)T(i+2)]$$

Where Ws represent weights and $T(i)$ represents the maximal epoch value ($T(2)$ in the previous expression) for the current minute, $T(i-1)$ for the previous minute, $T(i+1)$ for the succeeding minute, etc. Again, the computer varied the weighting and compared the resulting sleep/wake score with the EEG score until maximal agreement was obtained.

Seventeen of the 20 records were used in the algorithm development phase described above. The remaining three records were scored prospectively, i.e., each of the three records was scored individually with the single weighting and scale factor found optimal in the development phase. This test simulated the actual deployment of an automatic sleep scoring system, with the results compared with EEG scoring.

Results

The optimal algorithm reached after analysis of the 17 records was:

$$D = .025 \times [(.15T(i-4) + .15T(i-3) + .15T(i-2) + .08T(i-1) + .21T(i) + .12T(i+1) + .13T(i+2).]$$

where $T(i)$ represents the maximal epoch value in minute i, etc. If $D \geq 1.0$, the minute was scored *wake;* otherwise it was scored *sleep*. The best retrospective agreement between sleep/wake scored automatically with this algorithm and sleep/wake scored from EEG records was 94.46%—that is, 94.46% of all minutes in the file of 17 subjects were in agreement with the *true* sleep/wake score. Again, it should be noted that this is retrospective agreement, as the data for these individuals were used to select the optimal algorithm.

The ability of this algorithm to score sleep/wake prospectively was tested with the remaining three records. For these records, only the single expression found optimal in the algorithm development phase was used for automatic scoring of sleep/wake. Overall agreement of these three records with EEG scoring was 96.02%.

In order to understand the remaining shortcomings of the automatic sleep/wake scoring algorithm, data for all minutes misscored were listed and compared with the paper record. In general, the conditional probability of misscoring wake as sleep was higher (.062) than misscoring sleep as wake (.039). This difference is due in part to the fact that sleep onset is somewhat delayed after all activity ceases. Another factor contributing to the misscoring of wake as sleep was the presence of brief periods of inactivity during wakefulness. To correct for systematic context errors of this nature, a final stage of the sleep recognition algorithm was developed.

In this final stage the sleep/wake score for each minute as determined by the automatic scoring algorithm was subjected to additional conditions. These conditions took the form "after at least x minutes scored wake, the first y minutes scored sleep are rescored wake" and "y or less minutes scored sleep surrounded by at least x minutes scored wake (before and after) are rescored wake". After

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substituting a range of x and y values into these expressions and rescoring the records scored retrospectively, there emerged a combination of rules that increased still further the similarity between EEG and activity scoring. These rules were: (a) after at least 4 min scored wake, the first period of 1 min scored sleep is rescored wake; (b) after at least 10 min scored wake, the first 3 min scored sleep are rescored wake; (c) after at least 15 min scored wake, the first 4 min scored sleep are rescored wake; (d) 6 min or less scored sleep surrounded by at least 10 min (before and after) scored wake are rescored wake; and (e) 10 min or less scored sleep surrounded by at least 20 min (before and after) scored wake are rescored wake.

This rescoring procedure increased the minute-by-minute agreement between automatically scored activity data and EEG data for the 17 retrospectively scored records from 94.46% to 94.74%. More important for purposes of estimating total sleep time, this correction reduced the overestimation of the proportion of a record scored sleep by activity relative to EEG from 1.89% to 0.81%. A difference of 0.81% represents an overestimation of 11.65 min per 24-h day.

A further test conducted with these data estimated the resolution in the stored data necessary to achieve these levels of accuracy. The digital activity value was originally stored on disc as a 16-bit value, i.e., a number in the range of 0 to 32,767. To investigate the resolution requirement, the sleep recognition program was repeated with the same data, but the resolution of the data was reduced by dividing by powers of 2 and truncating the remainder. There were no decreases in agreement with 4-bit data (0−15) and a decrease of only 0.1% was found with 3-bit data (0−7). This surprising result is important, since it means that data for more minutes can be stored in a given memory capacity.

EXPERIMENT III

The outcome of Experiments I and II established the feasibility of a digital activity-based sleep monitor system capable of quantifying sleep time with reasonable accuracy while avoiding the impracticality and expense of polygraphic recording and scoring. Having developed the concepts in an experimental laboratory situation, we next desired to implement such a system in a form suitable for use in more naturalistic situations outside of the laboratory. To accomplish this purpose, the digitizing, preprocessing, and recording functions (generally those functions developed in Experiment I) were consolidated into a microprocessor-based digital monitor, small enough and light enough to be worn comfortably on a belt. The digital activity data collected by the monitor can be transferred to a larger computer for sleep/wake scoring (as in Experiment II).

Methods

The digital recorder was an early version of the PMS-8 monitor (Vitalog Corporation)[2] modified by Vitalog Corporation according to our specifications. It

[2] Vitalog Corporation, 1058 California Avenue, Palo Alto, CA 94306.

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consists of a IM6100 microprocessor, 8-channel A/D, 6K × 12 random access memory, crystal clock, and LED indicator. The entire package, including batteries, is enclosed in a plastic case 15 × 9 × 5.5 cm, and weighs 480 g. Communication to and from the monitor is provided by an interface to an Apple microcomputer. The monitor is fully programmable and capable of processing up to 8 analog inputs in real time, storing the digital data, and later transferring it to an Apple computer for permanent storage on diskettes.

The activity monitor program used for this study is derived from the algorithm found optimal in Experiment I. An external activity transducer, worn on a wrist, provides input to the monitor's A/D converter, and the analog signal is digitized at 240 Hz. The sum of four such conversions (representing the 60-Hz filter) is entered into a differencing function, equivalent to algorithm 5 in Experiment I. The output of the differencing function is summed for 2 s, and the largest 2-s score in each minute, scaled to 11 bits (0-2047), is stored sequentially in memory. Further, each minute a time code can be signaled through the LED. When coupled to a receiving photocell, this time code was written onto polygraph records to allow comparison of EEG and digital activity records. This monitor program collects digital activity data for 47 h 28 min.

Fourteen digital activity records totalling 12,739 min were collected from healthy male college students participating in experiments in which EEG, EOG, and EMG were being recorded. Time codes from the monitor LED were written on the polygraph record each minute. Since the physical parameters of the transducer-amplifier-A/D differed somewhat from those used previously, the parameters of the sleep recognition algorithm were recalibrated by the method described in Experiment II. For these recordings, the parameters producing the best agreement were:

$$D = .036 \times [.07T(i-5) + .08T(i-4) + .10T(i-3) + .11T(i-2) + .12T(i-1) + .14T(i) + .09T(i+1) + .09T(i+2) + .09T(i+3) + .10T(i+4)]$$

where $T(i)$ represents the greatest 2-s epoch score in minute i, etc., and $D \geq 1.0$ is scored wake. The results of this primary scoring algorithm were then subjected to the rescoring algorithm described in Experiment II, in which corrections for sleep onset latency and brief periods of inactivity were made. Overall minute-by-minute agreement was 93.88%. The activity estimate exceeded the EEG estimate of the proportion of sleep in a record by 1.03%, corresponding to 14.81 min per 24-h day. The minute-by-minute correlation coefficient between EEG scores and automatic estimates was 0.9724.

An additional 14 records totalling 22,514 min were scored prospectively by use of the parameters found optimal in the calibration phase. Twelve subjects were healthy male college students, and two were patients with sleep complaints who were undergoing clinical sleep evaluations. The patient records were included to test the system on highly disturbed sleep records. Overall agreement with EEG scoring was 93.04%. The activity estimate exceeded the EEG estimate of the proportion of sleep in a record by 1.20%, corresponding to 17.27 min per 24-h day. The correlation coefficient was 0.9692.

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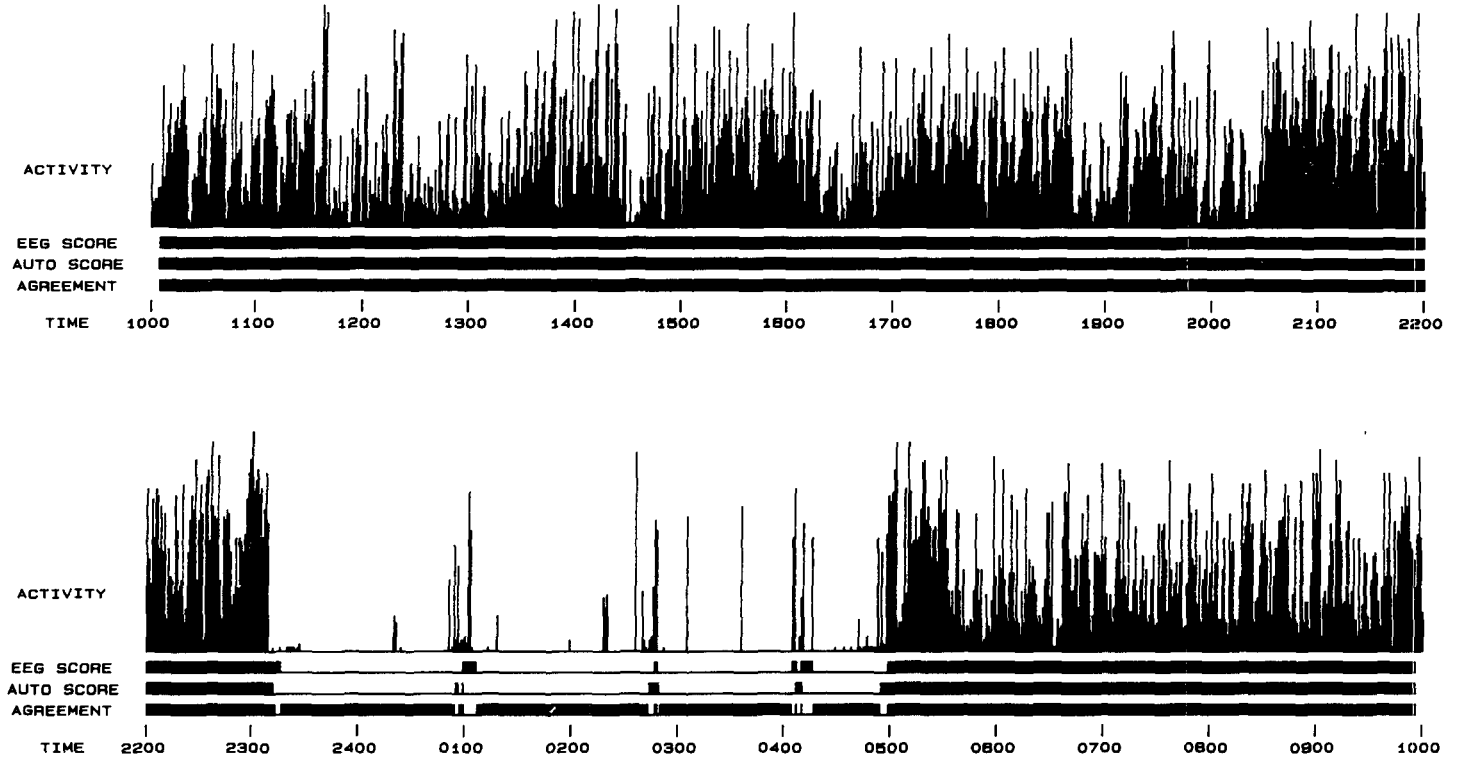


FIG. 3. Twenty-four-hour activity record collected with the digital monitor system divided into two 12-h segments. The top trace in each segment plots the digital activity value for the minute. The next trace is the result of EEG sleep/wake scoring (thick line represents wake, thin line sleep), and the automatic sleep/wake score derived from the activity data. The lowest trace illustrates the agreement between EEG and automatic scoring (thick line represents agreement).

Figure 3 illustrates a digital activity record in the prospective series. The 24-h record is divided into two 12-h segments, with the time marked along the bottom of each segment. The difference in activity level between sleep and wake and the close agreement between EEG and automatic sleep/wake scoring are apparent in this figure.

DISCUSSION

Mullaney et al. (1) have shown that a trained scorer can score wrist activity data for sleep/wake with accuracy approaching EEG scoring. The present study demonstrated that wrist activity data can be digitized and scored automatically by computer with little loss in accuracy and that the concept of digital activity recording can be implemented in a form suitable for ambulatory use. Mullaney et al. (1) estimated that their activity scoring system was 5 to 10 times less costly than EEG scoring, and the marginal decrease in accuracy is more than compensated for by the greater amount of data that can be collected with a given expense. We feel that the automatic scoring system described here further improves the cost-benefit relationship by replacing the largely mechanical analog recording and playback system, including the polygraph, with an all-digital system. Automatic scoring is accomplished in seconds, eliminating the time necessary for writing out a polygraph record and visually scoring the record. Elimination of a scorer further reduces costs and for the first time makes the identification of sleep and wake fully objective, without the many opportunities for error and variability presented by human scoring.

Many investigators assume that EEG scoring corresponds more accurately to behavioral and functional sleep than does activity scoring. Although we have no objective comparisons of EEG and activity scoring as indicators of sleep, we suspect that where the two methods are discrepant, behavioral criteria would on some occasions favor activity scoring. Given the known unreliability in hand scoring, some of the small discrepancies between activity and EEG scoring in the present study are undoubtedly due to inaccuracies in EEG scoring.

Even if the EEG method is slightly more accurate in estimating sleep time, other factors, such as simplicity and practicality, make activity scoring more desirable for many purposes. The activity transducer is more comfortable and convenient for the subject, who can sleep outside the laboratory if desired, free from electrodes. The adaptability of the activity recording and scoring method to naturalistic settings also increases its value for clinical and investigative measurement.

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